Chapter 2 : Linear Momentum

A. Linear Momentum of a particle

A particle of mass m and moving at a velocity $\vec{V}$, has a momentum $\vec{P}$ equal to the product of the particle's mass m by its velocity $\vec{V}$: $\vec{P} = m.\vec{V}$

- $\vec{V}$ is the instantaneous velocity of the paticle (at an instant t) and $\vec{P}$ is its momentum at that considered instant t.
- At a given position of the particle, $\vec{P}$, like $\vec{V}$, is tangent to the particle's trajectory at that position and has same direction as the movement. Its magnitude is: $P = m.V$
- m in kg, V in m/s, then P in kg.m/s

B. Linear momentum of a system of particles

For a system (S) constituted of particles A, B, C, ... of respective masses m_1 ; m_2 ; m_3; ... moving at respectives velocities $\vec{V}_1$, $\vec{V}_2$, $\vec{V}_3$, … . The linear momentum of the system is the sum of the linear momenta of its particles: $\vec{P}_S = \vec{P}_1 + \vec{P}_2 + \vec{P}_3 + \cdots = m_1\vec{V}_1 + m_2\vec{V}_2 + m_3\vec{V}_3 + \cdots$

C. Linear momentum of the center of mass of a system of particles

$\vec{P}_S = M_G.\vec{V}_G = \vec{P}_G$ Where G is the center of mass (or center of inertia or center of gravity) of the system.

Indeed: relative to a fixed point O taken as the origin of a frame of reference $(O\,;\vec{\imath}\,;\vec{\jmath}\,)$, the center of mass G of a system of particles A_1, A_2 , A_3 …. is given by: $\overrightarrow{OG} = \frac{m_1\overrightarrow{OA}_1 + m_2\overrightarrow{OA}_2 + m_3\overrightarrow{OA}_3 + \cdots}{m_1 + m_2 + m_3 + \cdots}$

So $(m_1 + m_2 + m_3 + \cdots).\overrightarrow{OG} = m_1\overrightarrow{OA}_1 + m_2\overrightarrow{OA}_2 + m_3\overrightarrow{OA}_3 + \cdots$; let $m_1 + m_2 + m_3 + \cdots = M_S$

Then $M_S.\overrightarrow{OG} = m_1\overrightarrow{OA}_1 + m_2\overrightarrow{OA}_2 + m_3\overrightarrow{OA}_3 + \ldots$ (1)

Lets derive (1) relative to t : $(M_S.\overrightarrow{OG})' = (m_1\overrightarrow{OA}_1)' + (m_2\overrightarrow{OA}_2)' + m_3(\overrightarrow{OA}_3)'$

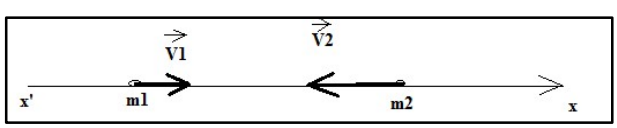
Then $M_S.(\overrightarrow{OG})' = m_1(\overrightarrow{OA}_1)' + m_2(\overrightarrow{OA}_2)' + m_3(\overrightarrow{OA}_3)'$ since the masses are considered constant

Finally : $M_S.\vec{V}_G = m_1\vec{V}_1 + m_2\vec{V}_2 + m_3\vec{V}_3$ Or $M_S.\vec{V}_G = \vec{P}_S$

As result, **the linear momentum of a system of particles of constant mass is equal to that of its center of mass affected by the total mass of the system:** $\boxed{\vec{P}_S = \vec{P}_G = M_G.\vec{V}_G = M_S.\vec{V}_G}$

Example1: A_1 (m_1 = 1Kg ; $\vec{V}_1 = 2\vec{\imath}\,(m/s)$) A_2 (m_2 = 500g ; $\vec{V}_2 = -4\vec{\imath}\,(m/s)$)

Determine $\vec{V}_G$. Conclude.



$\vec{P}_S = \vec{P}_1 + \vec{P}_2 = m_1\vec{V}_1 + m_2\vec{V}_2 = 1 \times 2\vec{\imath} + 0.5 \times (-4\vec{\imath}) = 0.\vec{\imath}$; then $\vec{P}_S = \vec{0}$

$\vec{V}_G$ =?

$\vec{P}_S = M.\vec{V}_G = \vec{0}$ then $\vec{V}_G = \vec{0}$: *The center of gravity of the system does not move; it keeps its position*.

Example 2: A_1 (m_1 = 1Kg ; $\vec{V}_1 = 2\cdot\vec{\imath} - 3\cdot\vec{\jmath}$) A_2 (m_2 = 200g ; $\vec{V}_2 = -3\cdot\vec{\imath} + 4\cdot\vec{\jmath}$)

$\vec{V}_G = ?$ $V_G = ?$

$$\vec{P}_S = \vec{P}_1 + \vec{P}_2 = m_1\vec{V}_1 + m_2\vec{V}_2 = 1\times(2\cdot\vec{\imath} - 3\cdot\vec{\jmath}) + 0.2\times(-3\cdot\vec{\imath} + 4\cdot\vec{\jmath}) = 2\vec{\imath} - 0.6\vec{\imath} - 3\vec{\jmath} + 0.8\vec{\jmath}$$
$$\vec{P}_S = 1.4\vec{\imath} - 2.2\vec{\jmath}$$

$$\vec{V}_G = \frac{\vec{P}_S}{M} = \frac{\vec{P}_S}{m_1+m_2} = \frac{1.4\vec{\imath}-2.2\vec{\jmath}}{1+0.2} = \frac{1,4}{1.2}\vec{\imath} - \frac{2,2}{1.2}\vec{\jmath} = 1.17\vec{\imath} - 1.83\vec{\jmath}\ ;$$

$$V_G = \sqrt{{V_x}^2 + {V_y}^2} = \sqrt{(1.17)^2 + (-1.83)^2} = 2.17m/s$$

D. General form of Newton's 2nd law

For a system (S), deformable or rigid, $\vec{P}_S = M_G.\vec{V}_G = M_S.\vec{V}_G$

Let's derive the above equation relative to time:

$\frac{d\vec{P}_S}{dt} = \frac{d(M_S.\vec{V}_G)}{dt} = M_G\cdot\frac{d\vec{V}_G}{dt}$ since $(\vec{V}_G)' = \vec{a}_G$ then $\boxed{\frac{d\vec{P}_S}{dt} = M_G\cdot\vec{a}_G = \sum\vec{F}_{ext}}$

This is the general form of Newton's 2nd law or the theorem of linear momentum.

Particular cases:

a. If : $\sum\vec{F}_{ext} = \vec{0}$; the system is said to be ***isolated*** or ***pseudo-isolated***, then : $\frac{d\vec{P}_S}{dt} = \sum\vec{F}_{ext} = \vec{0}$

We say there is ***conservation of linear momentum*** $\Rightarrow \vec{P}_S = \overrightarrow{constant}$ (i.e. $\vec{P}_f = \vec{P}_i$)

b. If : $\sum\vec{F}_{ext} = \overrightarrow{constant} \neq \vec{0}$, then : $\frac{\Delta\vec{P}_S}{\Delta t} = \frac{d\vec{P}_S}{dt} = \sum\vec{F}_{ext} \Rightarrow \Delta\vec{P}_S = \sum\vec{F}_{ext}\cdot\Delta t \Rightarrow \vec{P}_f - \vec{P}_0 = \sum\vec{F}_{ext}\cdot\Delta t$

We say that ***linear momentum is not consereved*** $(i.e.\vec{P}_f \neq \vec{P}_i)$.

E. Applications on the non-conservation of linear momentum

1- Braking of a vehicle

A car of mass M moves on a rough horizontal road at a velocity $\vec{V}_0$. Suddenly the driver acts on the brakes at $t_0 = 0$. The car stops at time t_1. Frictional forces and braking force are represtend respectively by constant forces $\vec{f}$ and $\vec{f}$', parallel to the road and opposite to the movement.

a. Express t_1 interm of f, f', M and V_0

b. Express the traveled distance D in term of M,V_0 , f and f'

Solution :

a. $t_1 = ?$

$$\Delta\vec{P}_S = \sum\vec{F}_{ext}\cdot\Delta t \Rightarrow \vec{P}_f - \vec{P}_0 = (M.\vec{g} + \vec{N} + \vec{f} + \vec{f}').(t_1 - t_0) \Rightarrow M.\vec{V}_f - M.\vec{V}_0 = (\vec{0} + \vec{f} + \vec{f}').t_1$$

By projections on $\overrightarrow{x'x}$ positively oriented in the direction of mvt:

$0 - MV_0 = (-f - f').t_1$ Finally: $\boxed{t_1 = \frac{-MV_0}{-f-f'} = \frac{MV_0}{f+f'}}$

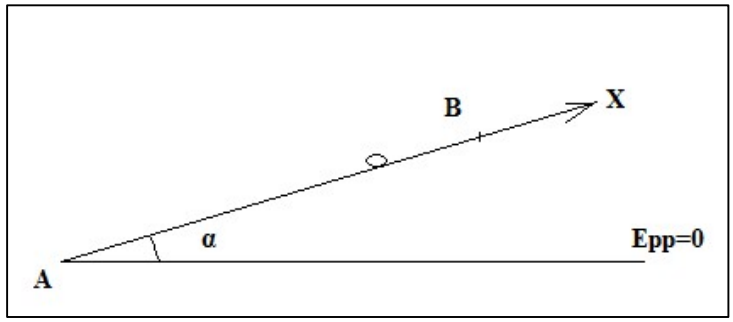
b. Non conservation of ME (existence of breaking force and friction)

$$\Delta ME = W_{\vec{f}} + W_{\vec{f}'} \Rightarrow \Delta KE + \Delta GPE = -f.D - f'.D$$

$$\Rightarrow \frac{1}{2}MV_f^2 - \frac{1}{2}MV_0^2 = -(f+f')D \Rightarrow \boxed{D = \frac{MV_0^2}{2(f+f')}}$$

2- Translation on an inclined plane

A ball of mass m = 500g is thrown on an inclined plane of α = 0.1 rad relative to the horizontal with a speed $V_0 = V_A = 3m/s$. The ball reaches point B with a speed $V_B = 1m/s$ during Δt =1.2s



a. Applying the general form of Newton's second law, determine the intensity of the frictional force.

System: Ball of center of mass G

$\sum \vec{F}_{ext} = m.\vec{g} + \vec{N} + \vec{f} = m.\vec{g}_x + m.\vec{g}_y + \vec{N} + \vec{f} = m.\vec{g}_x + \vec{f} \neq 0$; as result the ball system is not isolated and its linear momentum is not conserved. By the general form of newton's 2nd law:

$$\Delta\vec{P} = \sum \vec{F}_{ext} \cdot \Delta t \Rightarrow \vec{P}_B - \vec{P}_A = \left(m.\vec{g}_x + \vec{f}\right) \cdot \Delta t$$

By projection on $\overrightarrow{x'x}$: $mV_B - mV_A = (-mgsin(\alpha) - f) \cdot \Delta t \Rightarrow f = -mgsin(\alpha) - \frac{m(V_B - V_A)}{\Delta t} = 0.33N$

b. Determine the distance D traveled by the ball.

Friction $\neq 0$ then ME is not conserved: $\Delta ME_{A\rightarrow B} = W_{\vec{f}_{A\rightarrow B}} \Rightarrow ME_B - ME_A = -f.D$

$$\Rightarrow \left(\frac{1}{2}mV_B^2 + mgh_B\right) - \left(\frac{1}{2}mV_A^2 + mgh_A\right) = -f \cdot D \Rightarrow \frac{1}{2}m(V_B^2 - V_A^2) + mgDsin(\alpha) = -f \cdot D$$

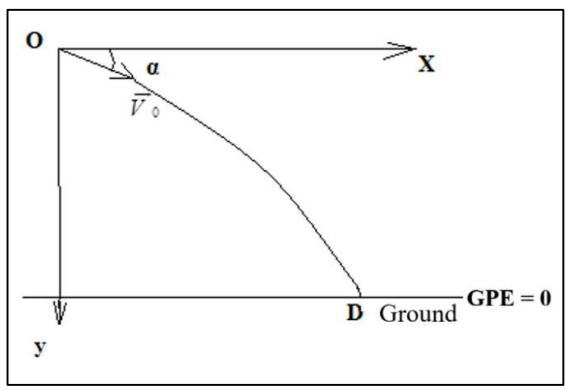
Finally: $D = \frac{m(V_A^2 - V_B^2)}{2(mgsin(\alpha) + f)} = 2.4\ m$

c. Calculate the amount of heat produced

$$Q = \Delta U = -\Delta ME = -W_{\vec{f}} = f \cdot D = 0.33 \times 2.4 = 0.792\ J = 0.792 \div 4.18 = 0.189\ cal$$

3- Projectile shot with an oblique velocity $\vec{V}_0$

A projectile of mass m = 100 g is thrown from a point O, with a speed V_0 = 10 m/s at 30° with the horizontal, as shown in the figure. Friction is supposed to be negligible. Given: h = 1m



a. Determine the speed of the projectile at point D just before reaching the ground.

No friction then ME is conserved:

$ME_D = ME_O \Rightarrow KE_D + GPE_D = KE_O + GPE_O$

$$\Rightarrow \frac{1}{2}mV_D^2 + 0 = \frac{1}{2}mV_O^2 + mgh$$

$$\Rightarrow V_D = \sqrt{V_O^2 + 2gh} = \sqrt{100 + 2 \times 10 \times 1} = 10.95\ m/s$$

b. Determine the P_x and P_y components of the momentum vector at time t.

System: Projectile of center of gravity G

$\sum \vec{F}_{ext} = m.\vec{g} \neq 0$; as result the ball system is not isolated and its linear momentum is not conserved. By the general form of newton's 2nd law:

$$\Delta\vec{P} = \sum \vec{F}_{ext} \cdot \Delta t \Rightarrow \boxed{\vec{P}_t - \vec{P}_O = m\vec{g} \cdot \Delta t}$$

- Projection on $\overrightarrow{Ox}$:
  $P_x - P_{0x} = 0 \Rightarrow P_x = P_{0x} = m \cdot V_{0x} = m \cdot V_0 \cdot \cos(\alpha) = 0.1 \times 10 \times \cos(30°) = 0.866\ kg.m/s$

- Projection on $\overrightarrow{Oy}$:
  $P_y - P_{0y} = mgt \Rightarrow P_y = mgt + P_{0y} = mgt + mV_0 \cdot \sin(\alpha) = t + 0.5$; then $\boxed{P_y = t + 0.5}$

c. Deduce the expressions of V_x and V_y at the instant t.

$$V_x = \frac{P_x}{m} = \frac{0.866\ kg.m/s}{0.1\ kg} = 8.66\ m/s$$
$$V_y = \frac{P_y}{m} = \frac{t + 0.5}{0.1} = 10t + 5$$

d. Determine the time equations of motion x (t) and y (t)

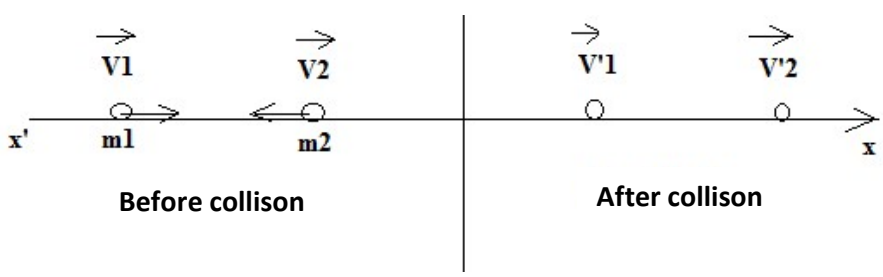
$$V_x = 8.66 \Rightarrow by\ anti - derivative: x = 8.66t + x_0 \Rightarrow x = 8.66t$$
$$V_y = 10t + 5 \Rightarrow by\ anti - derivative: y = 5t^2 + 5t + y_0 \Rightarrow y = 5t^2 + 5t$$

e. At what moment does the projectile reach the ground in D?

At point D, $V_D = 10.95$ m/s and $V_x = 8.66$ m/s
Or $V_D^2 = V_x^2 + V_y^2$ then $V_y^2 = V_D^2 - V_x^2 \Rightarrow Vy = \sqrt{10.95^2 - (8.66)^2} = 6.70\ m/s$ (Vy>0)
So: $10t + 5 = 6.70 \Rightarrow t = 0.17\ s$

F. Applications on the conservation of linear momentum.

1- Collision between two particles moving at colinear velocities



- Collision ⇒ conservation of total linear momentum

$\Rightarrow \vec{P}$(before collision) = $\vec{P}$(after collision) $\Rightarrow \vec{P}_1 + \vec{P}_2 = \vec{P'}_1 + \vec{P'}_2 \Rightarrow m_1\vec{V}_1 + m_2\vec{V}_2 = m_1\vec{V'}_1 + m_2\vec{V'}_2$.

Since the velocities are collinear, we can write algebraically:

$$m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2' \Rightarrow m_1(V_1 - V_1') = m_2(V_2' - V_2) \qquad \textbf{(1)}$$

- Elastic collision ⇒ conservation of total kinetic energy
  ⇒ $KE_{(Before\ collision)} = KE_{(After\ collision)} \Rightarrow KE_1 + KE_2 = KE_1' + KE_2'$

$$\Rightarrow \frac{1}{2}m_1V_1^2 + \frac{1}{2}m_2V_2^2 = \frac{1}{2}m_1V_1'^2 + \frac{1}{2}m_2V_2'^2 \Rightarrow m_1(V_1^2 - V_1'^2) = m_2(V_2'^2 - V_2^2)$$

$\Rightarrow m_1(V_1 - V_1')(V_1 + V_1') = m_2(V_2' - V_2)(V_2' + V_2)$ **(2)**

- $\frac{(2)}{(1)} \Rightarrow V_1 + V_1' = V_2 + V_2' \Rightarrow \boxed{V_2' = V_1 + V_1' - V_2}$ **(3)**
- **(3) in (1)** $\Rightarrow m_1(V_1 - V_1') = m_2(V_1 + V_1' - V_2 - V_2) \Rightarrow (m_1V_1 - m_1V_1') = m_2V_1 + m_2V_1' - m_2V_2 - m_2V_2$

  $\Rightarrow m_1V_1 - m_2V_1 + 2m_2V_2 = (m_1 + m_2)V_1' \Rightarrow \boxed{V_1' = \frac{(m_1-m_2)V_1+2m_2V_2}{(m_1+m_2)}}$ **(4)**
- **(4) in (3)** $\Rightarrow \boxed{V_2' = \frac{(m_2-m_1)V_2+2m_1V_1}{(m_1+m_2)}}$

Particular case When $V_2 = 0$, then $V_1' = \frac{(m_1-m_2)V_1}{(m_1+m_2)}$ and $V_2' = \frac{2m_1V_1}{(m_1+m_2)}$; as result :

- If $m_1 < m_2$, then
  $\frac{(m_1-m_2)}{(m_1+m_2)} < 0 \Rightarrow \vec{V'}_1$ and $\vec{V}_1$ are colinear but of opposite directions;
  $\frac{m_1V_1}{(m_1+m_2)} > 0 \Rightarrow \vec{V'}_2$ and $\vec{V}_1$ are colinear and of same direction.
  } *S_1 bounces off S_2 et S_2 starts moving in S_1 original direction*
- If $m_1 > m_2$, then
  $\frac{(m_1-m_2)}{(m_1+m_2)} > 0 \Rightarrow \vec{V'}_1$ and $\vec{V}_1$ are colinear and of same directions; similarly $\vec{V'}_2$ and $\vec{V}_1$ are colinear and of same direction. *S_1 continues moving in the same direction*.
- If $m_1 = m_2$, then
  $V_1' = 0$ and $V_2' = V_1$. *S_1 stops moving and S_2 moves at $\vec{V}_1$*

2- Colliding objects stick together

Collision $\Rightarrow$ conservation of total linear momentum

$\Rightarrow \vec{P}$(before collision) = $\vec{P}$(after collision) $\Rightarrow \vec{P}_1 + \vec{P}_2 = \vec{P}_S \Rightarrow m_1\vec{V}_1 + m_2\vec{V}_2 = (m_1 + m_2)\vec{V}_S$; as result

$$\boxed{\vec{V}_S = \frac{m_1\vec{V}_1 + m_2\vec{V}_2}{(m_1 + m_2)}}$$

3- Explosion

Collision $\Rightarrow$ conservation of total linear momentum

$\Rightarrow \vec{P}$(before collision) = $\vec{P}$(after collision) $\Rightarrow \vec{P}_S = \vec{P}_1 + \vec{P}_2 \Rightarrow \boxed{M_S\vec{V}_S = m_1\vec{V}_1 + m_2\vec{V}_2}$